MIL-T-46098A
29 September 1967

SUPERSEDING
MIL-T-46098(MR)
23 July 1965

MILITARY SPECIFICATION

TILE, CERAMIC, ALUMINA, HIGH STRENGTH

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high strength alumina ceramic tiles for lightweight composite armor applications.

1.2 Classification. The ceramic tile shall be of the following classes, as specified:

Class 1 - 93 percent minimum alumina
Class 2 - 83 percent minimum alumina

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9350

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:50Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS:

C18-60 - Methods for Chemical Analysis of Refractory Materials
C20-46 - Methods of Test for Apparent Porosity, Water Absorption, Apparent Specific Gravity and Bulk Density of Burned Refractory Brick
C328-56 - Flexural Properties of Fired Dry-Pressed Whiteware Specimens at Normal Temperature
E18-65 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
E-2 SM10-14-64 - Method for Spectrographical Chemical Analysis of Alumina Materials by the Powder D. C. Arc Technique

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Chemical composition and specific gravity. The chemical composition and specific gravity of the tile shall be within the limits established by the supplier, except that the minimum requirements shall be as specified in table I.

3.2 Mechanical properties. The minimum requirements for flexural strength and hardness shall be as shown in table I.

Table I. Minimum requirements for ceramic tile

	Class 1	Class 2
Alumina content	93%	83%
Specific gravity	3.58	3.37
Flexural strength ksi		
average	25	23
single value	22	20
Hardness		
Rockwell 45N or	76	73
Rockwell 30N	84	81

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:50Z
Check the source to verify that this is the current version before use.

3.3 Dimensions and tolerances. The dimensions of the tiles shall be as specified in the contract or the order (see 6.2). Unless otherwise specified in the contract or order, dimensional tolerances shall be as follows:

Thickness	Up to .355 thickness ± 0.005 inch Greater than .355 inch ± 0.008 inch
Length and width	± 0.005 inch
Angular	± 1/4°
Flatness	0.006 inch/inch

3.4 Identification marking. Each tile shall be ink-stamped or otherwise coded to provide identification with recorded data pertaining to manufacture and tests. The marking shall include the manufacturers identification. The materials and procedures used in marking shall not impair the materials properties (see 6.3).

3.5 Workmanship. Ceramic tiles shall be uniform in color and texture, and free from defects such as cracks, porosity, and fissures. Pits shall not be greater than 1/64 inch deep. Edge chips shall not exceed 1/32 inch in width or depth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. For the purpose of sampling a lot may consist of all tiles of a single size made from one ceramic batch and supplied in any one order but shall not exceed 10,000 tiles. For the purpose of this specification, the identity of the ceramic batch shall be established during the mixing process and shall be maintained during all subsequent operations. If a periodic kiln is used, each kiln load shall be identified separately.

4.3 Sampling.

4.3.1 For chemical analysis, hardness testing, specific gravity and flexural strength. One tile per thousand, but not less than four tiles shall be taken at random from each lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:50Z
Check the source to verify that this is the current version before use.

4.3.2 For examination. Sampling shall be in accordance with general inspection level II of MIL-STD-105. Acceptable quality levels shall be as specified in table II.

Table II. Sampling for examination

Requirement	Acceptable quality level percent defective
Dimensional tolerances	2.5
Flatness	2.5
Visual defects	2.5
Total defects per lot	1% of the number of tiles in the lot

4.4 Examination.

4.4.1 Visual. Each sample shall be visually inspected to determine compliance with the workmanship requirements.

4.4.2 Dimensional. Each sample shall be measured to determine compliance with the requirements for dimensions and tolerances.

4.5 Tests.

4.5.1 Chemical analysis. At lease one chemical analysis shall be made from each of the samples selected in 4.3.1. Chemical analysis, including determination of alumina content, shall be in accordance with ASTM Designation C18-60, or ASTM Method E-2 SM10-14-64. ASTM Designation C18-60 shall be used for referee analysis.

4.5.2 Flexural test. One specimen each from the samples selected in accordance with 4.3.1 shall be tested for flexural strength. Except as specified below tests shall be conducted in accordance with ASTM Designation C328-56.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:50Z
Check the source to verify that this is the current version before use.

4.5.2.1 Unless otherwise specified or permitted:

(a) Specimens shall be one-half inch in width and of the same thickness as the tile.
(b) Specimen surfaces containing original tile surfaces shall not be ground.
(c) The direction of loading shall be perpendicular to original tile surface.
(d) Three or four point loading may be used.

4.5.3 Specific gravity tests. One specimen each from the samples selected in 4.3.1 shall be tested for specific gravity. Tests shall be conducted in accordance with ASTM Designation C20-46.

4.5.4 Hardness tests. Rockwell 45N or Rockwell 30N hardness tests shall be conducted on samples selected in 4.3.1 in accordance with ASTM Designation E18-65.

4.5.5 Rejection criteria. Failure of any test specimen to meet the test requirements of 4.3.1 and 4.3.2 shall reject the lot represented.

5. PREPARATION FOR DELIVERY

5.1 Level C - Packaging. The high strength alumina ceramic tile shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Level C - Packing. The high strength alumina ceramic tile that requires overpacking by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery, and shall meet as a minimum, the requirements applicable to the mode of transportation.

5.3 Marking for shipment. In addition to any special marking specified in the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The high strength alumina ceramic tiles covered by this specification are intended for use in lightweight composite armor applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:50Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Class required
(c) Dimensions required (see 3.3)
(d) Levels of packaging and packing if other than level C as well as special markings (see Section 5)
(e) When test specimens differing from those specified in 4.5.2 are permitted

6.3 The materials and procedures used for tile identification marking should be compatible with materials and techniques used in manufacture of the composite armor.

Custodians:
Army - MR
Navy - AS
Air Force - 11

Review interest:
Army - MR, MO
Navy - AS
Air Force - 11

User interest:
Army - AV
Navy - SH

Preparing activity:
Army - MR

Project No. 9350-0058

Review/User information is current as of the date of this document; draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:50Z
Check the source to verify that this is the current version before use.